Casey Hartman

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Education

Université de Genève, Faculté Des Sciences — *Genève, Switzerland*

Master of Science in Astrophysics; Exoplanet Track GPA: 5.44/6.00, Overall: 5.35/6.00 — *Sept. 2025 - Jun. 2027*

Tufts University, College of Arts and Sciences — *Medford, Massachusetts*

B.S. in Astrophysics, Minor in Mathematics; Major GPA: 3.93, Overall GPA: 3.71 — *May 2025*

Honors: *Magna Cum Laude*

Publications

- **Hartman, C.**, Bourrier, V., et al., *"Stellar Tides in High-Resolution Transit Spectroscopy with ANTARESS"* (working title), in prep.
- Bourrier, V., et al. (incl. **Hartman, C.**), *"ATREIDES: Characterizing the Companions of Exo-Neptunes with Gaia DR4"* (working title), in prep.
- Bourrier, V., et al. (incl. **Hartman, C.**), *"ATREIDES: A Gaia DR4 Search for Wide-Orbit Companions of Exo-Neptunes"* (working title), in prep.

Relevant Coursework

Astronomy — Dynamics of Planetary Systems, Planet Formation & Evolution, Detection & Characterization Techniques, Planetary Atmospheres, Stars and Planets, Galaxies and Cosmology, Physical Processes in Astrophysics (Macro/Microscopic), Computational Astrophysics

Research Experience

Observatoire de Genève — *Genève, Switzerland*

MSc Thesis: Gaia Companions of Exo-Neptunes - Prof. Vincent Bourrier & Dr. João Faria — *Jun. 2026 - Present*

- Contribute to two ATREIDES (Bourrier et al. 2025) studies using Gaia DR4 epoch astrometry: characterizing known companions of exo-Neptunes, and searching for new ones over a broad range of masses and periods
- Compiled a homogeneous catalog of stellar, planetary, and companion properties for 420 Neptune systems, and cleaned all public and collaboration radial velocity data
- Built tools to jointly fit radial velocities and astrometry to find wide-orbit companions, tested on Hipparcos-Gaia proper motion anomalies ahead of the DR4 release
- Developing statistical methods to test how spin-orbit angles correlate with planet and companion properties, to constrain each Neptune's migration history

Stellar Tides in Transit Spectroscopy - Prof. Vincent Bourrier & Dr. Adrien Deline — *Jun. 2025 - Present*

- Extending ANTARESS (Bourrier et al. 2024), a public workflow for high-resolution transit spectroscopy, to model host stars distorted by both rotation and tides raised by a close-in planet, starting with WASP-12
- Built a tidal model based on Kaula (1964), transformed it into the ANTARESS frame, and combined it with the rotational potential to solve for the stellar surface at each orbital phase
- Calculated local surface gravity, gravity darkening, and limb darkening across the distorted surface
- Developing a numerical solver for the surface velocity field, since a tidally distorted photosphere has no closed-form line-of-sight intersection

Astrophysics Laboratory (MSc Semester Research Projects) — *Genève, Switzerland*

Self-Supervised Learning on JWST Spectra - Dr. Romain Meyer & Prof. Pascal Oesch — *Feb. 2026 - Jul. 2026*

- Adapted SimCLR to 1D spectra and trained on ∼20,000 NIRSpec prism spectra from the DAWN JWST Archive
- Without redshift labels, the network sorted spectra by redshift by learning the Lyman and Balmer breaks; redshifts estimated from its clusters matched known values with $R^2 = 0.855$
- Identified two candidate high-redshift analogs of GN-z11, and traced over-clustering of red sources to the data representation rather than the network

Disk Truncation in Binaries - Arianna Nigioni & Prof. Julia Venturini — *Sept. 2025 - Dec. 2025*

- Built a tool to compute circumprimary and circumsecondary truncation radii from binary eccentricity, separation, mass ratio, and disk viscosity, using the fits of Manara et al. (2019)
- Grew planetary embryos with a binary-adapted version of the Bern Model (Emsenhuber et al. 2021) in truncated circumprimary disks at separations of 50 to 300 AU
- Found that tighter binaries produce smaller planets that accrete less gas and migrate more slowly, while at 300 AU planets form nearly as they would around a single star

Tufts Department of Physics and Astronomy *Medford, Massachusetts*

Circumbinary Planet Search in TESS - Prof. David Martin *Sept. 2024 - Sept. 2025*

- Adapted STANLEY (Martin et al. 2021) to search TESS eclipsing binaries for circumbinary (P-type) planets
- Tested detrending methods on short-cadence, high-variability TESS light curves to find which removed systematics while preserving transits
- Validated the pipeline by recovering TOI-1338b, the first TESS circumbinary planet (Kostov et al. 2020)
- Added linear interpolation to the period vs. θ search grid, cutting compute time in half

Weizmann Institute of Science *Rehovot, Israel*

Particle Physics Research Intern - Prof. Ran Budnik *Jun. 2023 - Aug. 2023*

- Worked in the Experimental Astro-Particle Physics Lab on XENONnT, an international direct dark matter detection experiment
- Assisted in the calibration of a key component of XENONnT, a neutron generator, by designing an experiment to measure neutron emission rates at different distances and angles
- Analyzed experimental data in Python using statistical tests to characterize detected signals and calculate the neutron generator's flux

Teaching & Other Experience

Tufts Department of Physics and Astronomy *Medford, Massachusetts*

Teaching Assistant *Sept. 2022 - May 2025*

- Held bi-weekly office hours for students in the Astronomy, Mechanics, and Electricity and Magnetism classes
- Assisted in the creation and execution of lesson plans; proctored and graded homework and exams

Lucinetic *Pasadena, California*

Development and Data Science Intern *Jun. 2024 - Jan. 2025*

- Developed Python functions enabling Luci, the company's AI assistant for ArcGIS Pro, to search for, evaluate, and download non-terrestrial (Martian, Titan, etc.) datasets
- Built site selection demos with Luci in ArcGIS Pro and validated them against successful franchise locations

Extracurricular Activities

Students for the Exploration and Development of Space (SEDS) *Medford, Massachusetts*

Chair of the Board *May 2024 - May 2025*

- Responsible for overall management of SEDS (100+ active members), including organizing and leading the board, project teams, and general club meetings; Liaison between other Tufts clubs and SEDS' chapters

Core Member *Sept. 2021 - May 2025*

- Helped design and build a small radio telescope on the physics building roof, including wind loading calculations
- Founder of astrophotography team; Travel to dark sites to take photos of extra-terrestrial objects

Director of Finance *May 2022 - Apr. 2024*

- Increased the SEDS operating budget from $8,000 to $50,000; tracked, monitored, and approved expenses

Achievements & Presentations

2025	**Departmental Seminar Presentation**	*Université de Genève*
2025	**SPS Zone Meeting Presentation**	*Tufts University*
2025	**Dean's List (8x)**	*Tufts University*
2019	**Eagle Scout**	*Los Angeles, California*

Skills

Software	Python, LaTeX, Git, Docker, Django, DACE, ArcGIS Pro, Fusion 360, Microsoft Office Programs
Libraries	Matplotlib, Numpy, Pandas, Scipy, Astropy, arvi, Wotan, Lightkurve, Rebound, Keras, Selenium, etc.
Telescope Data	TESS, Kepler/K2, Gaia, Hipparcos, JWST, Various Spectrographs incl. ESPRESSO

Interests

Philanthropy	Space Kids (Founder), Camp Harmony Counselor, Robotics Coach, SOVA Food Pantries, Red Cross
Interests	Piano, Astrophotography, Reading, Pickleball, Skiing, Baseball, Rock Climbing
Organizations	AtomicMind (Tutor), Bad Neighbours (Rock Band), Society of Physics Students, Tufts Ski Racing & Rock Climbing